Microscopic Features of the Oral Cavity (Set 2)

1. Histologically, the oral cavity is lined by stratified squamous epithelium. Regions subjected to constant mechanical friction during mastication (such as the gingiva) feature which specific type of epithelium?
a. Simple columnar epithelium
b. Pseudostratified ciliated columnar epithelium
c. Keratinized (or parakeratinized) stratified squamous epithelium
d. Non-keratinized stratified squamous epithelium
e. Transitional epithelium
2. The mucosal lining of regions like the lips, lips-cheek transitions, and the floor of the mouth is not subjected to severe friction and is classified histologically as:
a. Masticatory mucosa
b. Specialized mucosa
c. Lining mucosa (non-keratinized stratified squamous)
d. Respiratory mucosa
e. Gustatory epithelium
3. Masticatory mucosa is characterized histologically by the presence of a dense lamina propria that attaches directly to the underlying bone, a structural relationship termed a:
a. Submucosal fat cushion
b. Mucoperiosteum
c. Elastic ligament sheet
d. Synovial lining matrix
e. Chondroid plate
4. Microscopic evaluation of parakeratinized stratified squamous epithelium reveals that the cells in the superficial stratum corneum retain their:
a. Mitotic spindles
b. Intercalated discs
c. Pyknotic nuclei
d. Desmosomal junctions completely intact
e. Mucous secretory vacuoles
5. Microscopic analysis of the hard gingival tissue reveals a distinct superficial stratum corneum, beneath which is a prominent layer containing dark keratohyalin granules, known as the:
a. Stratum basale
b. Stratum spinosum
c. Stratum granulosum
d. Stratum lucidum
e. Lamina propria
6. The cells of the stratum spinosum (prickle cell layer) are bound tightly to each other by numerous intercellular bridges. Under high magnification, these bridges correspond to:
a. Gap junctions
b. Tight junctions (zonula occludens)
c. Desmosomes (macula adherens)
d. Focal adhesions

e. Hemidesmosomes

7. The epithelial layer of the oral mucosa is anchored to the underlying connective tissue lamina propria by a specialized extracellular matrix structure called the:

a. Reticular capsule
b. Basement membrane (basal lamina)
c. Synovial barrier
d. Perichondrium
e. Periodontal ligament

8. Basal epithelial cells attach onto the lamina lucida of the basement membrane via cellular junctions known as:

a. Desmosomes
b. Tight junctions
c. Hemidesmosomes
d. Gap junctions
e. Maculae adherens

9. The connective tissue lamina propria features finger-like projections that interlock with epithelial ridges (rete pegs) to prevent mechanical shearing of the mucosa. These projections are:

a. Submucosal villi
b. Dermal ridges
c. Dermal papillae (connective tissue papillae)
d. Fibrous trabeculae
e. Keratin spurs

10. Scattered within the basal layers of the oral stratified squamous epithelium are clear dendritic cells responsible for synthesizing protective pigment granules. These are:

a. Langerhans cells
b. Merkel cells
c. Melanocytes
d. Keratinocytes
e. Fibroblasts

11. Microscopic evaluation of human gingiva reveals a specialized epithelial zone that faces the tooth surface and attaches directly to the enamel matrix to form a biological seal. This zone is the:

a. Sulcular epithelium
b. Junctional epithelium
c. Free gingival epithelium
d. Alveolar mucosa epithelium
e. Keratinized masticatory sheet

12. Junctional epithelium cells are non-keratinized and maintain a unique structural design, featuring exceptionally wide intercellular spaces that allow for the regular migration of:

a. Ameloblasts
b. Odontoblasts
c. Polymorphonuclear leukocytes (white blood cells) and gingival fluid
d. Enamel rods
e. Osteocytes

13. The sublingual space mucosa features a thin, non-keratinized epithelium overlying a highly vascularized lamina propria. This design makes the sublingual mucosa a vital systemic route for the rapid absorption of:
a. Keratin scales
b. Insoluble food debris
c. Sublingual nitroglycerin tablets for angina patients
d. Salivary zymogen polymers
e. Extracellular cementum anchors
14. The microscopic cells responsible for sensing touch, pressure, and thermal variations located within the connective tissue papillae of the oral mucosa are:
a. Pacinian corpuscles and Meissner's corpuscles
b. Langerhans dendritic clusters
c. Type II gustatory light units
d. Basal stem cells
e. Melanocyte clusters
15. Pathological evaluation of an oral mucosal biopsy reveals localized loss of the epithelial cell layer exposing the underlying inflamed lamina propria core. This tissue destruction is defined as an:
a. Hyperplasia mass
b. Oral ulcer
c. Hyperkeratosis patch
d. Acanthosis fold
e. Sialolith block
16. The major salivary glands are high-yield exocrine systems. Histologically, the secretory parenchyma of the parotid gland is composed exclusively of:
a. Mucous acini with serous demilunes
b. Pure serous acini
c. Pure mucous acini
d. Cord-like solid endocrine columns
e. Stratified squamous nests
17. Microscopic evaluation of a serous acinus reveals a ring of pie-shaped cells surrounding a small lumen. The spherical nuclei of these cells are characteristically located in which position?
a. Pressed flat against the basal cell membrane
b. Suspended at the absolute apical plasma border
c. Centrally or within the basal third of the cell
d. Extruding into the canalicular lumen
e. Displaced completely outside the cytoplasm matrix
18. The cytoplasm of serous acinar cells is highly unique under the microscope, featuring an abundance of rough endoplasmic reticulum basally and packed secretory structures apically called:
a. Mucus vacuoles
b. Zymogen granules
c. Lipid droplets
d. Glycogen pools
e. Keratohyalin granules

19. The submandibular salivary gland is a mixed gland. Histologically, it is predominantly serous but features a small percentage of mucous acini capped by crescent-shaped groups of serous cells called:
a. Crypts of Lieberkühn
b. Serous demilunes (of Giannuzzi)
c. Hassall's corpuscles
d. Malpighian layers
e. Von Ebner caps
20. Microscopic evaluation of a mucous acinus reveals cells filled with large, pale-staining secretory droplets. The nuclei of mucous cells are characteristically:
a. Perfectly spherical and central
b. Undergoing active mitotic division
c. Flattened and compressed against the basal cell membrane
d. Fragmented into pyknotic strands
e. Suspended in the apical cytoplasm
21. The sublingual salivary gland is also a mixed gland, but unlike the submandibular gland, its secretory parenchyma is predominantly:
a. Pure serous
b. Mucous (often with serous demilunes)
c. Fibrous connective tissue scar pads
d. Reticuloendothelial cells
e. Squamous cell nests
22. Embracing the outer basal surfaces of salivary acini and intercalated ducts are specialized contractile epithelial cells that squeeze the acinus to expel secreted saliva into the duct system. These are:
a. Myoepithelial cells (basket cells)
b. Plasma cells
c. Fibroblasts
d. Mast cells
e. Chondrocytes
23. The duct system of major salivary glands is highly branched. The smallest ducts that lead directly out of the acinar lumen are lined by a single layer of low cuboidal cells and are termed:
a. Striated ducts
b. Intercalated ducts
c. Interlobular ducts
d. Main excretory ducts
e. Wharton's tracts
24. Intercalated ducts lead into striated ducts. Striated ducts are lined by simple columnar epithelium and display prominent basal striations under high magnification, which are produced by:
a. Dense parallel rows of keratin filaments
b. Thick bundles of skeletal actin lines
c. Infoldings of the basal plasma membrane packed with vertically oriented mitochondria
d. Intercalated cell junctions
e. Ciliary axonemes

25. Striated ducts are highly active metabolically. Their primary histological function is to modify the primary saliva fluid by reabsorbing sodium and chloride ions while secreting:
a. Zymogen enzymes
b. Mucus droplets
c. Potassium and bicarbonate ions
d. Ameloblast matrix
e. Calcium phosphate scales
26. Interlobular and excretory salivary ducts run within the connective tissue septa separating gland lobules. Their epithelial lining gradually transitions from simple columnar to:
a. Stratified squamous non-keratinized epithelium
b. Simple squamous sheets
c. Pseudostratified columnar or stratified cuboidal/columnar epithelium
d. Keratinized scales
e. Transitional rows
27. The connective tissue capsule of major salivary glands extends inward as septa to divide the parenchyma into lobes and lobules. This stroma contains plasma cells that produce which protective antibody?
a. IgG
b. IgM
c. Secretory IgA (sIgA)
d. IgE
e. IgD
28. Bone tissue of the alveolar process supports the teeth. Under the microscope, the bone that lines the tooth socket and receives the insertion of the periodontal ligament fibers is called:
a. Spongy bone
b. Cancellous trabecular matrix
c. Bundle bone (or alveolar bone proper/cribriform plate)
d. Cortical plate bone
e. Woven bone callus
29. The dense bundles of collagen fibers belonging to the periodontal ligament that are embedded directly into the bundle bone and cementum matrices are named:
a. Purkinje fibers
b. Sharpey's fibers
c. Elastic fibers of Tome
d. von Korff's fibers
e. Perforating Haversian tracks
30. The hard tissue covering the anatomical root of a tooth is cementum. Histologically, acellular cementum covers the upper coronal half of the root, while cellular cementum is located apically and contains trapped:
a. Ameloblasts
b. Odontoblasts
c. Cementocytes inside lacunae
d. Osteoclasts
e. Fibroblasts

31. The core substance of a tooth is dentin, which is a vital mineralized avascular tissue. Dentin is traversed by thousands of microscopic channels stretching from the pulp to the outer perimeter, called:
a. Haversian canals
b. Volkmann's canals
c. Dentinal tubules
d. Canaliculi of bone
e. Enamel spindles
32. Inside each dentinal tubule runs a cytoplasmic extension (Tomes' fiber) belonging to the cells that form dentin, which line the outer pulp boundary. These cells are:
a. Ameloblasts
b. Odontoblasts
c. Cementoblasts
d. Osteoblasts
e. Fibroblasts
33. The dental pulp is the central soft tissue of the tooth. Under the microscope, it represents a specialized loose connective tissue containing a rich vascular network, nerves, and four distinct layers, starting with the outer:
a. Odontogenic zone (odontoblast layer)
b. Cell-free zone (of Weil)
c. Cell-rich zone
d. Pulp core central zone
e. Parietal plexus of Raschkow
34. The highly unique, heavily mineralized hard tissue covering the anatomical crown of a tooth is dental enamel. Enamel is completely unique because it lacks any organic cells or collagen and is composed of millions of:
a. Osteons
b. Enamel rods (prisms) and interrod substance
c. Cementicles
d. Dentinal matrices
e. Calcified chondrocyte rings
35. Enamel is formed embryologically by ameloblasts, which are completely lost when the tooth erupts. Consequently, dental enamel can never undergo:
a. Physical wear or attrition
b. Acid demineralization
c. Cellular repair or regeneration
d. X-ray visibility changes
e. Orthodontic movement shift
36. Maxillary and mandibular permanent teeth have different structural variations under the microscope. The incremental lines that appear as concentric brown rings in cross-sections of enamel, representing evidence of rhythmic mineral deposition, are:
a. Lines of Owen
b. Striae of Retzius
c. Perikymata grooves
d. von Ebner lines
e. Hunter-Schreger bands
37. The alternating dark and light bands seen in longitudinal sections of enamel due to regular directions changes in adjacent groups of rods are:

a. Lines of Owen
b. Contour lines of Retzius
c. Hunter-Schreger bands
d. Tomes' granular layers
e. Neonatal lines

38. The hypomineralized structural variations that project from the dentinoenamel junction (DEJ) into the enamel short distances, resembling tiny shrubs or brushes, are:

a. Enamel tufts
b. Enamel lamellae
c. Enamel spindles
d. Dentinal tubules
e. Interglobular spaces

39. Hypomineralized leaf-like structural defects that extend from the enamel surface all the way down to the DEJ, serving as potential pathways for caries bacteria, are:

a. Enamel tufts
b. Enamel lamellae
c. Enamel spindles
d. Interglobular dentin fields
e. Incremental line tracks

40. Odontoblast processes that occasionally cross the DEJ into the early enamel matrix and become trapped as blind, club-shaped structures are called:

a. Enamel tufts
b. Enamel lamellae
c. Enamel spindles
d. Tomes' fibers
e. Neonatal tracks

41. Dentin matrix that is newly secreted by odontoblasts and remains unmineralized until a later phase is formally designated as:

a. Intertubular dentin
b. Peritubular dentin
c. Mantle dentin
d. Predentin
e. Globular dentin

42. The first layer of dentin formed in the tooth crown, located immediately deep to the enamel, is characterized by slightly different mineral configurations and is termed:

a. Circumpulpal dentin
b. Mantle dentin
c. Predentin
d. Secondary dentin
e. Tertiary dentin

43. The bulk of the dentin in both the crown and root, located beneath the mantle dentin layer, is known as:

a. Interglobular dentin
b. Sclerotic dentin
c. Circumpulpal dentin
d. Predentin matrix
e. Secondary regular dentin

44. Dentin that is formed slowly and continuously throughout the life of a vital tooth after complete root formation has occurred is classified as:
a. Primary dentin
b. Secondary dentin
c. Tertiary (reparative) dentin
d. Sclerotic dentin
e. Interglobular dentin
45. Localized, irregular dentin matrix deposited rapidly at a specific site on the pulp wall in response to local injuries like dental caries or deep cavity preparations is:
a. Primary dentin
b. Secondary dentin
c. Tertiary (reparative or reactive) dentin
d. Mantle dentin
e. Predentin
46. Unmineralized or hypomineralized pockets of dentin matrix formed when regular calcified spheres fail to fuse normally in the circumpulpal zone are called:
a. Intertubular dentin lines
b. Peritubular tracks
c. Interglobular dentin
d. Tomes' granular layers
e. Sclerotic blocks
47. A distinct histological zone of hypomineralized dark spots seen exclusively in the radicular (root) dentin immediately deep to the cementum layer is:
a. The striae of Retzius
b. Tomes' granular layer
c. The lines of von Ebner
d. The contour lines of Owen
e. Interglobular crown spaces
48. Microscopic evaluation of an old tooth reveals that many dentinal tubules have become completely filled with mineral, closing the lumen and making the dentin appear transparent. This tissue alteration is:
a. Dead tracts
b. Sclerotic (translucent) dentin
c. Interglobular dentin
d. Predentin hyperplasia
e. Secondary pulp calcification
49. When severe trauma causes odontoblasts to die, their processes degenerate, leaving behind empty, air-filled dentinal tubules that look black under transmitted light. These are:
a. Sclerotic bands
b. Dead tracts
c. Interglobular fields
d. Mantle columns
e. Intertubular zones
50. The rich, dense plexus of unmyelinated nerve fibers located immediately deep to the odontoblast layer in the dental pulp is the:
a. Parietal plexus of Raschkow
b. Meissner's plexus

c. Auerbach's plexus
d. Haversian network
e. Volkmann's network

Answer Key Table

Q 1-8	Q 9-16	Q 17-24	Q 25-32	Q 33-40	Q 41-48	Q 49-50
c	**c**	**c**	**a**	**c**	**b**	**a**
d	**b**	**b**	**c**	**b**	**c**	**b**

b	**c**	**b**	**c**	**a**	**c**	
c	**a**	**c**	**c**	**c**	**b**	
c	**b**	**b**	**b**	**c**	**c**	
c	**b**	**b**	**b**	**c**	**b**	
b	**c**	**a**	**a**	**c**	**b**	
c	**b**	**c**	**b**	**a**	**a**	